

Sai Bharath Seelam

✉ saibharathseelam0399@gmail.com ☎ +1 (615) 810-1584 🌐 <https://www.linkedin.com/in/sbs10>

SUMMARY

Data Professional with **4 Years of Expertise in Analytics, Engineering, and AI-Powered Solutions** delivering transformative data-driven solutions across diverse industries, including e-commerce, supply chain, cybersecurity, and customer engagement. Adept at designing and deploying robust data architectures, scalable pipelines, and predictive models that drive operational excellence and business growth. Recognized for leveraging advanced machine learning techniques, cloud platforms, and statistical frameworks to extract actionable insights and optimize complex workflows.

Key highlights:

- **Predictive Analytics Expertise:** Designed and implemented machine learning frameworks such as XGBoost, Scikit-learn, and TensorFlow, achieving high accuracy and business value, including a **96.4% churn prediction accuracy** and a **27% reduction in inventory holding costs**.
- **Data Engineering Mastery:** Developed **real-time streaming pipelines with Kafka, Apache Spark, and Kubernetes**, processing millions of records with near-zero latency and ensuring 99.99% uptime for mission-critical systems.
- **Innovative Solutions Delivery:** Spearheaded AI-powered platforms such as supply chain optimization ecosystems and cybersecurity threat intelligence automation, leading to **\$450K in cost savings** and **65% faster breach response times**.
- **End-to-End Visualization and Reporting:** Designed interactive dashboards and reporting tools using **Tableau, Power BI, and Prometheus**, empowering data-driven decision-making with real-time insights.

SKILL SET

Data Engineering & Architecture

- ETL Development: Apache Airflow, AWS Glue, Apache NiFi, Apache Spark
- Cloud Platforms: AWS (S3, Redshift, SageMaker), Azure (Data Lake, Synapse Analytics), Google Cloud (BigQuery)
- Data Warehousing: Snowflake, Amazon Redshift, Azure Synapse Analytics
- Workflow Orchestration: Kubernetes, Terraform, Docker

Machine Learning & Analytics

- Machine Learning Libraries: Scikit-learn, TensorFlow, PyTorch, XGBoost
- Predictive Modeling: Time-Series Analysis, Neural Networks (LSTM, CNN), Ensemble Methods
- Feature Engineering & Optimization: NumPy, pandas, Vectorization, Statistical Feature Selection
- Statistical Analysis: Hypothesis Testing, Regression Analysis, Multivariate Analysis

Big Data & Real-Time Processing

- Streaming Data Platforms: Apache Kafka, Flink, Spark Streaming
- Distributed Computing: Hadoop, PySpark
- Data Lakes & Pipelines: Delta Lake, AWS Lake Formation

Visualization & Reporting

- Dashboard Development: Tableau, Power BI, Looker
- Real-Time Monitoring: Prometheus, Grafana
- Geospatial and Advanced Visualizations: Plotly, Matplotlib, Seaborn

Programming & Scripting

- Programming Languages: Python, SQL, R, Scala, Java
- Query Optimization: Advanced SQL (CTEs, Window Functions), NoSQL (MongoDB, Cassandra)
- Automation: Bash Scripting, Python Automation

PROFESSIONAL EXPERIENCE

Data Analyst

Dell Technologies

May 2023 – present | Remote, US

- **Predictive Customer Resilience Analytics Framework**
 - Engineered machine learning pipeline using **XGBoost and scikit-learn** to predict enterprise customer churn with **96.4% probabilistic accuracy**.
 - Implemented **advanced feature engineering** techniques, extracting nuanced signals from 75,000+ multidimensional customer interaction datasets.
 - Developed **real-time anomaly detection algorithm** using statistical inference, enabling proactive risk mitigation strategies.
 - Optimized data processing workflow, reducing computational complexity by **40%** through **vectorized NumPy transformations**.
- **Hyper-Personalized E-commerce Revenue Intelligence Platform**
 - Architected **multi-dimensional segmentation model** leveraging **probabilistic machine learning** to unlock granular consumer behavior insights.
 - Utilized **advanced statistical hypothesis testing** and **multivariate regression techniques** to deconstruct conversion rate dynamics.
 - Designed **automated ETL pipeline** using **Apache Spark** for processing 2M+ transaction records with near-zero latency.
 - Synthesized complex data patterns, generating **actionable recommendation engine** that drove **22% incremental revenue optimization**.
- **AI-Powered Supply Chain Optimization Ecosystem**

- Developed **neural network-based demand forecasting model** using TensorFlow, reducing inventory holding costs by **27%**.
 - Implemented **advanced time-series analysis** with **LSTM algorithms** to predict supply chain disruption risks.
 - Created **real-time predictive dashboard** using Power BI, integrating machine learning insights for strategic decision-making.
 - Engineered **automated anomaly detection system** that identified critical logistical inefficiencies, saving \$450K in operational expenses.
- **Cybersecurity Threat Intelligence Automation Platform**
 - Created **advanced anomaly detection algorithm** using **deep learning neural networks** for real-time security threat identification
 - Implemented **threat vector correlation system** reducing potential security breach response time by 65%
 - Utilized machine learning ensemble techniques to classify complex network intrusion patterns
 - Developed comprehensive threat intelligence reporting mechanism for enterprise security teams

Data Analyst

Jun 2020 – Dec 2021 | Hyderabad, India

AT&T

- **Advanced Sales Forecasting Using Predictive Analytics**
 - Engineered and deployed a robust **time-series forecasting model using Python, SQL, and ARIMA/XGBoost algorithms**, resulting in a **12% improvement in sales forecast accuracy** across a \$10M product portfolio.
 - Automated data pipelines via **Apache Airflow**, integrating with Google BigQuery for real-time reporting and ensuring 99.9% data consistency while reducing manual processing time by 35%.
 - Designed and implemented **feature engineering pipelines**, optimizing key predictors with advanced statistical techniques such as **Lag Features and Fourier Transforms**, enhancing model performance by 15%.
 - Integrated **AWS SageMaker** for model deployment, enabling seamless CI/CD workflows and reducing model retraining time by 40% while adhering to scalability requirements.
- **Customer Segmentation and Retention Strategy via Clustering Techniques**
 - Conducted **K-Means and Hierarchical Clustering** on 10M+ customer records, leveraging **ETL workflows and visualization tools like Tableau and Power BI** to identify 5 actionable customer segments that drove a **20% increase in targeted retention campaigns' ROI**.
 - Implemented **SQL-based cohort analysis and churn prediction models**, achieving a **30% reduction in churn rates within 6 months** by integrating insights into marketing workflows using APIs and CRM tools like Salesforce.
 - Leveraged **Python's Scikit-learn and NumPy libraries** to create custom data transformation pipelines, ensuring compatibility across machine learning models with a 25% boost in processing efficiency.
 - Developed **interactive dashboards in Tableau**, incorporating dynamic filtering and geospatial visualizations, empowering leadership teams with actionable insights that expedited decision-making cycles by 22%.

Data Engineer

Oct 2020 – Mar 2021 | Bangalore, India

MoneyGram

- **High-Performance Data Analytics Pipeline**
 - Engineered a **Kafka-based streaming data pipeline** capable of processing 1M+ events per minute, achieving 99.99% uptime and sub-second processing latency.
 - Designed a **partitioned data lake architecture in Azure Data Lake**, optimized for query performance and cost-efficiency, reducing storage expenses by 30%.
 - Leveraged **Airflow and Kubernetes** to orchestrate containerized workflows, enabling horizontal scaling and reducing pipeline processing time by 40%.
 - Built end-to-end monitoring solutions with **Prometheus and Grafana**, providing real-time alerts and visual dashboards, reducing incident response time by 35%.
- **Scalable Data Warehouse Architecture for Real-Time Analytics**
 - Designed and implemented a **cloud-native data warehouse using Amazon Redshift and Snowflake**, enabling real-time analytics on 5TB+ of daily transactional data, reducing query latency by 40%.
 - Developed and maintained robust **ETL pipelines using Apache NiFi and AWS Glue**, ensuring seamless data ingestion from 15+ sources with a 99.8% reliability rate.
 - Automated schema validation and anomaly detection using **PySpark and Delta Lake**, improving data integrity checks and reducing critical errors by 25%.
 - Deployed **Terraform** for infrastructure-as-code (IaC), enabling scalable and version-controlled deployment of cloud resources, reducing provisioning times by 50%.

EDUCATION

University of Memphis

Jan 2022 – May 2023 | Memphis, Tennessee

Master of Science in Information Systems

SR University

Apr 2017 – May 2021 | Telangana, India

Bachelor of Engineering in Electronics and Communication Engineering

CERTIFICATES

- Data Analytics for Management
- Business Project Management